

PROFILE

Final-year Data & AI Engineering student at ESILV with hands-on experience in data pipelines, applied machine learning, RAG systems and reliable AI backends. I build LLM applications where retrieval, validation and software controls protect business data. Seeking a **six-month Data Engineering, Applied AI or Machine Learning internship starting February 2027**.

PROFESSIONAL EXPERIENCE

Data & AI Engineering Intern

May 2026 to Sep 2026

B-AGILE SARL, Essor AI Copilot

Casablanca, Morocco

- Built an end-to-end ERP copilot with React, FastAPI, Pydantic, PostgreSQL and SQLAlchemy, covering purchasing, receptions, stock and supplier returns from sanitized business exports.
- Implemented semantic request planning, hybrid RAG with embeddings and cited sources, deterministic validation, and controlled actions with preview, confirmation, idempotency and audit trails.

Software Engineering Intern, AIOps & Machine Learning

Apr 2025 to Jun 2025

Société Générale ABS

Casablanca, Morocco

- Prepared operational incident data, engineered features and built scikit-learn components for anomaly detection and incident classification.
- Prototyped a RAG assistant over technical runbooks using document chunking, embeddings and semantic search to support incident diagnosis.

Data Analyst Intern

Jun 2025 to Aug 2025

A.G.T.T, Agence Générale de Transit et de Transports

Casablanca, Morocco

- Built Python and SQL ETL workflows to consolidate operational data and automate recurring Excel reports and data-quality checks.

SELECTED ENGINEERING PROJECTS

Essor AI Copilot, Controlled RAG Assistant for ERP

2026

Python · FastAPI · PostgreSQL · SQLAlchemy · Alembic · React · RAG · Docker

- Designed a semantic planner using an NVIDIA-compatible LLM with a deterministic fallback, mapping French requests to typed, allowlisted PostgreSQL operations instead of model-generated SQL.
- Built a hybrid RAG pipeline with chunking, metadata enrichment, embeddings, vector indexing, lexical and semantic search, RRF fusion, evidence thresholds and source citations; added safe confirmable writes.

AIOps Incident Intelligence & RAG Assistant, Société Générale ABS

2025

Python · FastAPI · PostgreSQL · Kafka · scikit-learn · NLP · RAG · Docker

- Structured incidents, API health checks and monitoring events in PostgreSQL, then exposed operational indicators through FastAPI services.
- Combined feature engineering, anomaly detection and classification with an embedding-based semantic search pipeline over runbooks to improve incident triage and diagnosis.

B2B Lead Qualification Engine, ESILV × Red Alert Labs

Oct 2025

Python · MongoDB · Web Scraping · JSON Schema · Airtable

- Built an end-to-end pipeline for public B2B lead data, with separate RAW and CLEANED MongoDB layers for traceability and auditability.
- Made ingestion idempotent and recoverable with schema validation, retries and backfills, while applying GDPR-conscious data processing.

EDUCATION

Engineering Degree in Data & Artificial Intelligence, Final Year

2025 to 2027

ESILV, De Vinci Engineering School

Paris, France

- Machine Learning, Deep Learning, NLP, Big Data, Cloud Computing, Data Engineering and statistical modeling.

Licence MIAGE, Computer Science Applied to Management

2022 to 2025

Université Internationale de Casablanca

Casablanca, Morocco

- Software engineering, Java, SQL, databases, information systems and project management.

TECHNICAL SKILLS

Programming and Data: Python, SQL, PostgreSQL, MongoDB, Pandas, ETL/ELT, data modeling, data quality and Kafka fundamentals.

AI and Machine Learning: scikit-learn, NLP, RAG, LLM orchestration, prompt engineering, embeddings, vector search, semantic search, classification, anomaly detection and SHAP.

Backend and Delivery: FastAPI, Pydantic, SQLAlchemy, Alembic, REST APIs, Docker, Git, GitHub Actions, Java and Spring Boot.

Analytics and Languages: Power BI, Tableau, Airtable, Plotly and Streamlit. Arabic: native; French and English: fluent.